

Escaping Temporary Traps?

Evidence from Age-Based Incentives to Permanent Employment

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Motivation: labor market duality

Open-ended (OE)

- Higher wages & severance
- Job stability
- Career progression

Short-term (ST)

- Lower wages, no severance
- High turnover
- Limited progression

Duality is pervasive across Europe and developing countries (Eichhorst et al., 2018; ILO, 2015).

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Standard view

Duality arises from the **cost gap** in employment protection between OE and ST (Blanchard and Landier, 2002).

⇒ Prescription: reduce the cost gap to lower ST prevalence.

At least 128 reforms in Europe since 1980 (Boeri, 2011)

This paper

Does cutting the OE–ST cost gap reduce duality — and *who benefits?*

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Setting: Spain's 1997 reform

- Payroll tax subsidies reduce OE non-wage costs by **9–12%** of gross wages
- ST costs **unchanged** $\Rightarrow$ clean shift in relative costs
- **Two age notches** (30 & 45)
- Firm-side eligibility restrictions generate within-age variation in subsidy access

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Why Spain?

- Highest ST incidence in OECD pre-reform: **33.6%** vs. 10.9% OECD avg. (1996)
- Large OE–ST cost gap in Europe: $\approx$ **6.4–7%** of ST labor cost (Bentolila et al., 2012)
- Two age thresholds $\Rightarrow$ young *and* older worker margins in one reform

Roadmap

Data and Institutional Context

Young Workers (18–30)

Older Workers (45+)

Wage Incidence

Elasticity of Substitution

Mechanisms

Conclusion

Data and Institutional Context

Data (MCVL)

- **Muestra Continua de Vidas Laborales (MCVL):** administrative longitudinal panel from Spanish Social Security records, population register, and tax administration.
- **Coverage:** 4% random sample of all individuals with Social Security activity in each year.
- **Variables:**
 - Employment and unemployment spells (daily resolution).
 - Contract type (OE/ST, low-cost vs. high-cost subsidy indicator).
 - Monthly wages, separations, hires.
 - Worker: age, gender, education, citizenship, disability.
 - Firm: size, sector, location, age, legal status.
- Retrospective histories: analyze the 1997 reform using MCVL waves 2004–2012.

The 1997 reform: design and a caveat

Cost reductions for new OE hires only

- **Payroll tax cuts:** 25% (ages 18–30) and 38% (>45) of payroll taxes $\approx$ 9–12% of gross wages
- **Lower severance** for wrongful dismissal: 45 $\rightarrow$ 33 days/year of tenure
- ST costs **unchanged** $\Rightarrow$ clean reduction in relative OE cost

Identification leverage

- Age notches at **30** and **45** $\Rightarrow$ sharp eligibility discontinuities
- Firm-side eligibility restrictions $\Rightarrow$ **low-cost vs. high-cost** OE variation *within* age groups $\Rightarrow$ internal control group

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But: statutory costs $\neq$ total hiring costs

$\Rightarrow$ Even when OE costs fall *below* ST, firms may retain ST due to **wage premia** demanded by workers and **legal uncertainty** around dismissal

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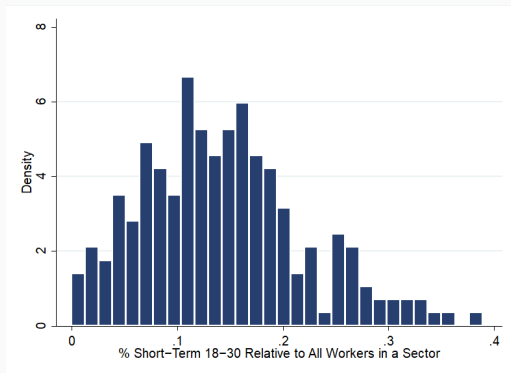
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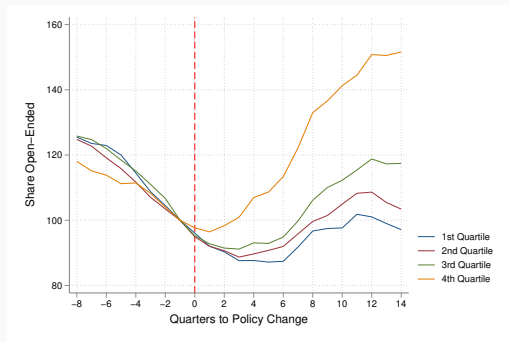
Identification: sectoral exposure among young workers

Intuition: sectors with more ST workers pre-reform face stronger incentives to convert.

Treatment intensity: $\%ST_{s,96}$ = ST share of 18–30 workers in sector s , 1996



(a) Distribution of $\%ST_{s,96}$



(b) OE employment share by quartile

Identification: estimating equation

Treatment: $D_s = \mathbb{1}[\%ST_{s,96} \in Q4]$ — sectors in the top quartile of pre-reform ST exposure vs. the bottom quartile.

$$Y_{st} = \delta_s + \delta_t + \sum_{j \neq -1} \beta_j \mathbb{1}[t=j] \times \%ST_{s,96} + \sum_j \gamma_j \mathbb{1}[t=j] \times X_s + \varepsilon_{st}$$

Y_{st} OE employment share in sector s , period t

δ_s Sector fixed effects

δ_t Time fixed effects

$D_s = \mathbb{1}[\%ST_{s,96} \in Q4]$: top-quartile ST exposure (vs. Q1)

X_s Sector-level controls (interacted with time)

$t = -1$ Omitted (reference) period

Young workers: transitions into open-ended contracts

	Coefficient	SE
From a Short-Term contract		
ST → Open-Ended (any)	2.891***	(0.915)
ST → Low-Cost OE	2.994***	(1.035)
ST → High-Cost OE	−0.104	(0.381)
Job-to-job mobility within OE		
Low- → High-Cost OE (different firm)	0.059**	(0.025)

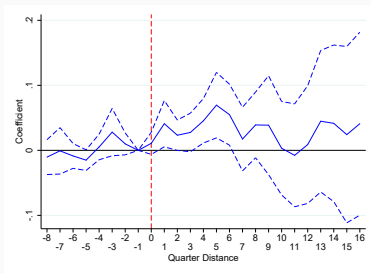
- ST→OE transitions rise **strongly**, almost entirely into *low-cost* OE.
- No effect on transitions into *high-cost* OE $\Rightarrow$ firms use the subsidy.
- Job-to-job moves to higher-cost OE *at different firms* $\uparrow \Rightarrow$ **stronger outside options**.

Young workers: employment stocks and entry from non-employment

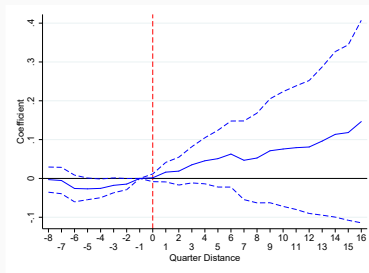
	Coefficient	SE
Panel A: Employment stocks		
OE (all)	0.170***	(0.052)
OE (low-cost)	0.093***	(0.027)
OE (high-cost)	0.074	(0.045)
Ever low-cost OE	0.111***	(0.030)
Short-Term	-0.034	(0.095)
ST employment share	-0.140***	(0.036)
Panel B: Transitions from Non-Employment		
NE → Low-Cost OE	0.109*	(0.055)
NE → Any OE	0.053	(0.093)
NE → Short-Term	0.026	(0.033)

- OE employment ↑ **17pp**; ST share ↓ **14pp**.
- Growth concentrated in low-cost OE; ST stock falls modestly.
- Entry from non-employment shifts toward low-cost OE — no overall employment expansion through new hires.

Spillovers to workers 30–65



(a) ST Employment 30–65



(b) OE Employment 30–65

- **No displacement** of adults from OE.
- ST $\uparrow$ among 30–65 in more-exposed sectors.
- **Interpretation:** when converted young workers move into core tasks, firms staff residual flexible tasks with older ST workers (often part-time/short duration).

Older Workers (45+)

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Older workers: two-step approach

1. **Near the notch (age 45):** quantify timing distortions (delayed conversions) and identify unaffected age ranges.
2. **Far from the notch:** compare 40–41 vs. 47–48 to estimate employment effects above eligibility.

Step 1: Notch analysis
(counterfactual density)
around age 45



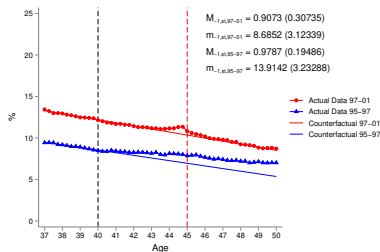
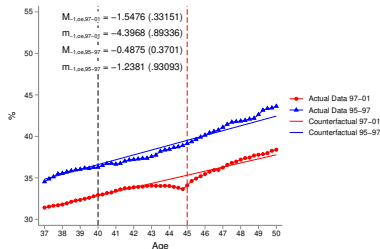
Step 2: Far-from-notch
DiD: 40–41 vs. 47–48

Step 1: Counterfactual density around age 45

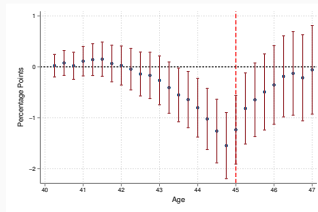
- Build counterfactual employment-by-age profile using ages *below* a_L (e.g., 37–40).
- Compare observed vs. counterfactual to measure:
 - delayed ST→OE conversions
 - across-notch substitution
 - crowd-out of high-cost OE
- Bootstrap-based inference.

$$c_{jk} = \beta_k a_j + \sum_{i \geq a_L} \gamma_{ik} \mathbb{1}[a_j = i] + v_{jk},$$

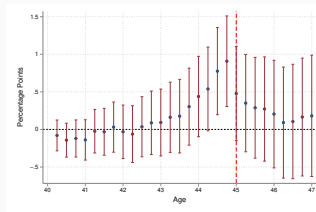
$$\hat{M}_{jk} = c_{jk} - \hat{c}_{jk}, \quad \hat{m}_{jk} = \frac{c_{jk} - \hat{c}_{jk}}{\hat{c}_{jk}}$$



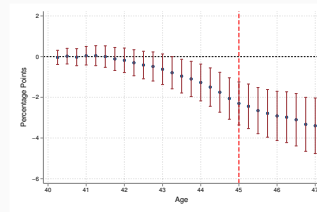
Step 1 results: timing distortions & crowd-out



(a) OE Employment



(b) ST Employment



(c) High-Cost OE

- ST $\uparrow$ just below 45 (delayed conversions dominate); OE $\downarrow$ then $\uparrow$.
- Limited across-notch substitution.
- **High-cost OE crowded out** near the notch by low-cost OE.
- Define unaffected ages $\rightarrow$ 40–41 vs. 47–48.

Step 2: Strategy far from the notch (40–41 vs. 47–48)

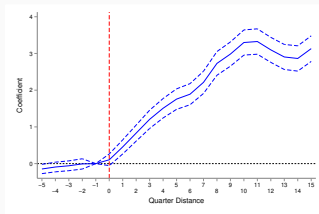
- Compare outcomes for ages 47–48 (eligible) vs. 40–41 (ineligible), away from local notch distortions.
- DiD / event study with rich controls and month FE:

$$y_{ijt} = \alpha + \eta \text{Above45}_i + \delta \text{post}_t + \beta \text{Above45}_i \times \text{post}_t + \gamma X_{it} + \rho_t + \varepsilon_{it}.$$

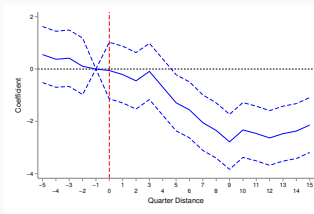
Outcomes of interest

- Transitions: ST→OE; non-employment→OE.
- Employment: total OE, low-cost OE, high-cost OE, ST.

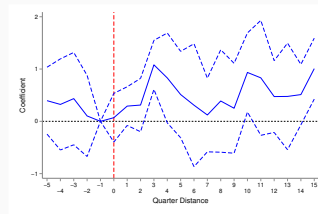
Step 2 results: no net OE expansion above 45



(a) Low-Cost OE



(b) High-Cost OE



(c) Total OE

- Low-cost OE $\uparrow$ **offset** by high-cost OE $\downarrow$; total OE $\approx$ unchanged.
- **Subsidy reallocates within OE; does not create new OE jobs.**

Wage Incidence

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Wage incidence: method

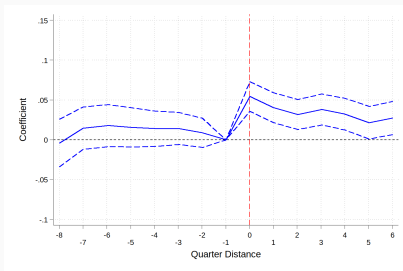
- Compare wage trajectories of ST→OE transitions into **low-cost** vs. **high-cost** OE.
- Individual FE + time FE + age FE; controls interacted with event time.

$$Y_{it} = \alpha + \delta_i + \delta_t + \delta_a + \sum_{j=-8, \neq -1}^{16} \beta_j \mathbb{1}[j = t] \text{Treatment}_i + \sum_{j=-8}^{16} \gamma_j \mathbb{1}[j = t] X_i + \rho X_{it} + \varepsilon_{it}.$$

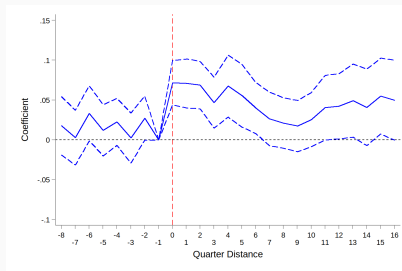
Logic

- Both groups transition to OE $\Rightarrow$ comparable selection on the transition margin.
- Difference in wages = differential cost reduction $\Rightarrow$ **pass-through**.

Wage incidence: event-study estimates



(a) Workers 18–30



(b) Workers 45–55

- Wages jump for low-cost relative to high-cost OE at conversion.
- Effects persistent over post-treatment horizon.

Elasticity of Substitution

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Definition

$$\sigma = - \frac{\Delta \ln(L_{OE}/L_{ST})}{\Delta \ln(W_{OE}/W_{ST})}.$$

- **Numerator:** relative employment change.
 - Young: $\Delta \ln(L_{OE}/L_{ST}) \approx 0.370$
 - Older: $\Delta \ln(L_{OE}/L_{ST}) \approx 0.0013$
- **Denominator:** relative cost change, accounting for wage pass-through ω .
 - Young: $\omega = 0.372 \Rightarrow$ effective OE cost cut $\approx 4.14\%$
 - Older: $\omega = 0.557 \Rightarrow$ effective OE cost cut $\approx 5.05\%$
 - Plus ST wage change: $+1.5\%$ young, 0 older.
 - $\Delta \ln(W_{OE}/W_{ST}) \approx 0.0488$ young; 0.0518 older.

Baseline

- Young (18–30): $\sigma \approx 7.6$ (95% CI [-2.18, 17.31])
- Older (45+): $\sigma \approx 0.025$ (95% CI [-0.06, 0.11])
- Near age-45 notch: $\sigma \approx 2.35$ (timing margin)

Sensitivity (range across specifications)

- Young: $\sigma \in [4.0, 8.7]$
- Older: $\sigma \in [0.01, 0.08]$
- **Interpretation:** ST and OE are close substitutes for young workers; effectively non-substitutable for older workers.

Mechanisms

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Why does σ differ so sharply by age?

- **Adverse selection:** older ST workers are more negatively selected on observables (wage, experience, occupation) than younger ST workers $\Rightarrow$ far from OE productivity threshold.
- **ST wage penalty:** ST jobs pay less even conditional on observables $\Rightarrow$ ST may persist despite shrinking statutory gap.
- **Task allocation:** ST may serve seasonal/replacement demand for older workers (Garibaldi and Gomes, 2022; Del Bono and Weber, 2008) $\Rightarrow$ contract types serve distinct functions.

Adverse selection table

ST wage penalty table

Mechanism 1: Adverse selection intensifies with age

	ST vs. High-Cost OE		Low- vs. High-Cost OE	
	Below 30	Above 45	Below 30	Above 45
Log Wage	−0.30***	−5.98***	−5.43***	−12.54***
Experience in OE	−0.29***	−0.61***	−0.32***	−0.45***
Experience in ST	−0.20***	−0.28***	−0.27***	−0.47***
High-Rank Occupation	−6.52***	−16.27***	−3.09***	−8.43***
University Degree	−2.31***	−3.10***	−5.49***	1.78
Firm Size	−0.35***	−1.68***	−0.32***	−1.32***
Observations	319,355	81,575	54,731	29,627

- Older ST workers are **much** more negatively selected than younger ST workers (wages, experience, occupation).
- Same pattern for low-cost OE hires: above-45 entrants further below the high-cost OE benchmark.
- Consistent with near-zero σ for older workers: too far from the OE productivity threshold.

Mechanism 2: ST wage penalty

- Cross-sectional conditional gap: $ST < OE$ wages, penalty $\approx 15.7\%$.
- Within-worker $ST \rightarrow OE$ wage gain: $\approx 9\%$.

Table 1: Wage Penalty for Short-Term Contracts

	(1)	(2)	(3)	(4)	(5)
	log wage				
ST	-0.196*** (0.002)	-0.191*** (0.002)	-0.122*** (0.002)	-0.100*** (0.003)	-0.157*** (0.007)
Observations	533,422	533,422	533,422	306,237	47,685

- ST and OE differ in more than statutory costs $\Rightarrow$ aligning labor costs is necessary but *not sufficient* to eliminate duality.

Conclusion

Conclusion

- Lowering OE costs can **break temporary traps** for young workers: large ST→OE substitution and stronger outside options.
- For older workers, **monetary incentives alone do not create marginal OE jobs**; low-cost OE mainly replaces high-cost OE.
- **Wage incidence**: substantial but incomplete pass-through, larger than predicted by tax-benefit linkage alone $\Rightarrow$ targeted demand shifts generate rent sharing.
- **Spillovers**: reforms reshape task allocation and contract composition beyond treated groups (adult ST $\uparrow$ in exposed sectors).
- **Policy implication**: cost incentives effective for youth duality, but need complementary tools (training, hiring norms) to address adverse selection and structural temporariness for mid-career workers.

Thank you!

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Appendix

Appendix: Adverse selection (full table)

Table 2: Sorting into Short-Term and Low-Cost Open-Ended Contracts

	(1) st	(2) st	(3) st	(4) bonified	(5) bonified	(6) bonified
Log Wage	-0.00336*** (0.000766)	-0.0577*** (0.00298)	-0.00316*** (0.000419)	-5.513*** (0.342)	-12.63*** (0.578)	-1.816*** (0.0307)
Experience in OE	-0.00297*** (0.0000469)	-0.00616*** (0.0000615)	-0.00585*** (0.0000117)	-0.315*** (0.0119)	-0.451*** (0.0120)	-0.110*** (0.000850)
Experience in ST	-0.00199*** (0.0000509)	-0.00275*** (0.0000959)	0.000455*** (0.0000196)	-0.261*** (0.0161)	-0.455*** (0.0221)	0.0532*** (0.00142)
High-Rank Occupation	-0.0699*** (0.00366)	-0.158*** (0.0125)	-0.0598*** (0.00132)	-3.144*** (1.065)	-7.478*** (1.576)	1.231*** (0.0956)
University Degree	-0.0223*** (0.00178)	-0.0338*** (0.0113)	-0.0114*** (0.000890)	-5.032*** (0.704)	0.795 (1.495)	0.0204 (0.0657)
Firm Size	-0.00386*** (0.000223)	-0.0183*** (0.000596)	-0.0150*** (0.0000933)	-0.483*** (0.0789)	-1.397*** (0.0757)	-0.109*** (0.00641)
Log Wage x 45			-0.0601*** (0.00308)			-11.42*** (0.560)
Experience in OE x 45			-0.000989***			-0.334***

Appendix: Short-term wage penalty (full table)

Table 3: Wage Penalty for Short-Term Contracts

	(1)	(2)	(3)	(4)	(5)
	log wage				
ST	-0.196*** (0.002)	-0.191*** (0.002)	-0.122*** (0.002)	-0.100*** (0.003)	-0.157*** (0.007)
Observations	533,422	533,422	533,422	306,237	47,685

Notes: Estimated wage penalty faced by short-term (ST) workers relative to OE workers. Sample: ages 18–55, four quarters before to six quarters after the reform. Column (1) excludes controls; Column (2) adds province and time FE; Column (3) is the preferred specification with full controls. Columns (4)–(5) restrict to ages 18–30 and 45–55. Robust SE in parentheses. * 10%, ** 5%, *** 1%.

Appendix: Pass-through from non-employment to OE

- Alternative entry margin: $NE \rightarrow OE$.
- Pass-through estimates: **108%** young; **79.5%** older (Table C5, col. 5).
- Caveats: smaller samples, positive selection.
- Qualitatively: confirms substantial pass-through for older workers.

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Appendix: Sensitivity of σ estimates

Table 4: Sensitivity Analysis of Elasticity Estimates

Sensitivity	σ
<i>Panel A: Young workers</i>	
Continuous treatment (average)	6.8
First 3 quarters only	6.6
No change in ST labor costs	8.7
No change in standard OE wages	6.3
No wage incidence ($\omega = 0$)	4.0
<i>Panel B: Older workers</i>	
75% of ST due to spillover	0.054
50% of ST due to spillover	0.082
No wage incidence ($\omega = 0$)	0.011

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